

Third-year CS undergraduate working on how automated systems make consequential decisions about people — and how to keep them auditable. Author of RuleWeaver, an Apache-2.0 compiler turning statutory prose into provenance-preserving executable rules. Rebuilt Priora, a prior-authorization system, around criterion-level evidence after watching a risk score start displacing the judgement it was meant to support. Has programmed 17 technical events reaching 500+ students across two campus communities, and writes weekly for 1,700+ followers on human-in-the-loop AI adoption and governance.

SELECTED WORK

RuleWeaver — Rules-as-Code compiler · Python, Apache-2.0

github.com/mozenjii/RuleWeaver

- Open-source compiler turning statutory prose into provenance-preserving, testable, executable rules — typed IR, closed expression AST, and a deterministic four-state evaluator emitting execution traces. 131 commits, 389 tests.
- Designed the extraction safety boundary: proposed rule extractions must clear six independent checks before entering the rule set — IR parse, cited-clause match, verbatim quote resolution, forced *needs_review* status, recorded-but-unused confidence, and prompt-injection escalation. Confidence is recorded and deliberately never consulted by the evaluator.
- Built a 41-diagnostic verification engine with type checker, boundary and date-transition generators, and a mutation harness catching 22/22 planted faults; semantic diff quantifies amendment impact on real household outcomes.

Priora — Provider-side prior-authorization control plane · Python

- Began as a denial-risk prediction prototype over structured and free-text inputs. Rearchitected after the risk score began organising the review queue — and, through it, shaping outcomes it was only meant to triage.
- Now evidence-first: criterion-level provenance, versioned policy, and a canonical state machine. Each determination emits which criterion was evaluated, against which policy version, on what cited evidence, and from where.
- Enforced structurally, not by convention — a lint rule forbids any denial-probability field from serving as a primary queue sort key or feeding any gate or validator. 33 product and 44 technical specification files, 344 machine-readable artifacts.

Epoques — AI Systems Architect · Remote

Nov 2025 – May 2026

- Built context-engineered LLM workflows with guardrails and vector retrieval for SMEs in healthcare, retail, and professional services, targeting a 25–35% cut in manual operational work.

COMMUNITY LEADERSHIP & OUTREACH

Microsoft Learn Student Ambassadors, Islamabad — Lead, Outreach & PR

Jan 2025 – Dec 2025

- Mobilised 10+ campus leads to host 5 Islamabad networking events (250+ attendees) and cross-campus skill workshops; secured 12 industry experts for keynotes and hands-on sessions, owning outreach and logistics.
- Ran targeted social campaigns that lifted event registrations 35% and widened ecosystem visibility.

Google Developer Groups on Campus, FAST — Head, Technical Workshops

Oct 2024 – May 2025

- Programmed and ran 12 technical seminars and webinars engaging 250+ students, building the campus workshop track from scratch.
- Curated and delivered sessions on physics–computing integration and Kolmogorov–Arnold Network research, translating frontier papers for mixed-skill student audiences and seeding three new interdisciplinary projects.

Algorizms — Co-Founder & CEO

Jun 2025 – Dec 2025

- Designed campus outreach and screening that recruited 23 interns from 600+ applicants — a repeatable student-recruiting funnel.
- Ran sprint cadences to ship a B2B SaaS MVP on time, secured 15+ pilot demos, lifted adoption 40%, and wrote the onboarding playbooks that cut contributor ramp-up.

Also: Vice Head, Code Craft — NaSCon 2025 · PR & sponsorships — Software Engineering Society, FAST · Outreach — FAST Quran & Sunnah Society · Founder & President, Coding & Development Society — LGS Paragon

WRITING & SPEAKING

Publish weekly to 1,700+ followers on where LLMs belong in real workflows — baselining before automating, designing for the exception path over extraction accuracy, and provenance under the EU AI Act and NIST GenAI profile.

TECHNICAL SKILLS

GenAI: LLM APIs, context engineering, guardrail & eval design, prompt-injection defence, provenance and human-review gating, vector retrieval, agentic workflows

AI/ML: Python, scikit-learn, pandas, NumPy, TF-IDF, supervised classification, mutation & property testing, pytest

Backend & data: Node.js, Express, SQL Server, REST APIs, schema design, data pipelines, Git, GitHub Actions

HONOURS

- 3rd place — National AI Hackathon (atomcamp), Lahore 2026, five-city national
- Silver Medalist, FAST NUCES · 2× Dean's List
- A-Levels A*AA & High Achiever's Award — LGS Paragon